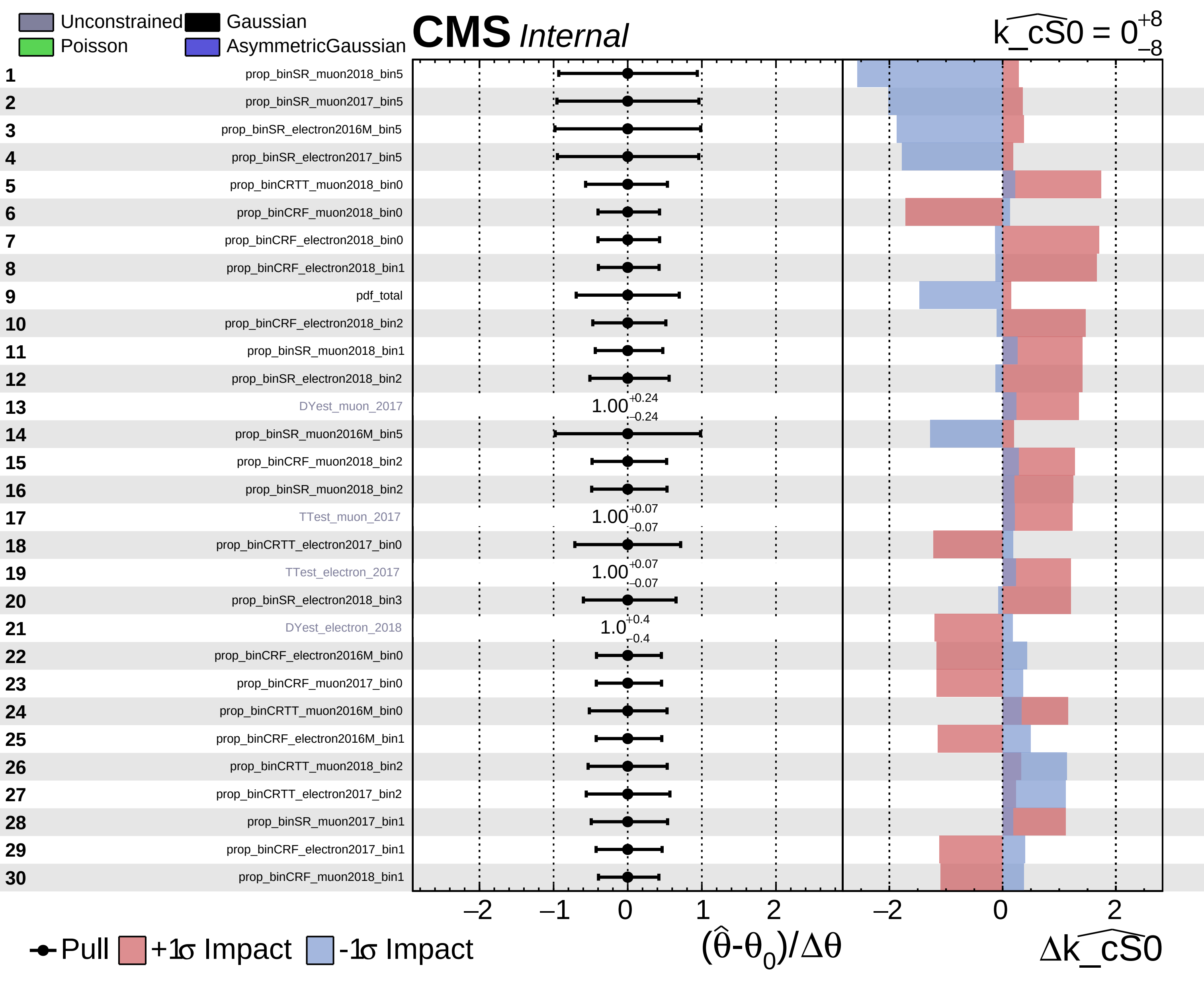
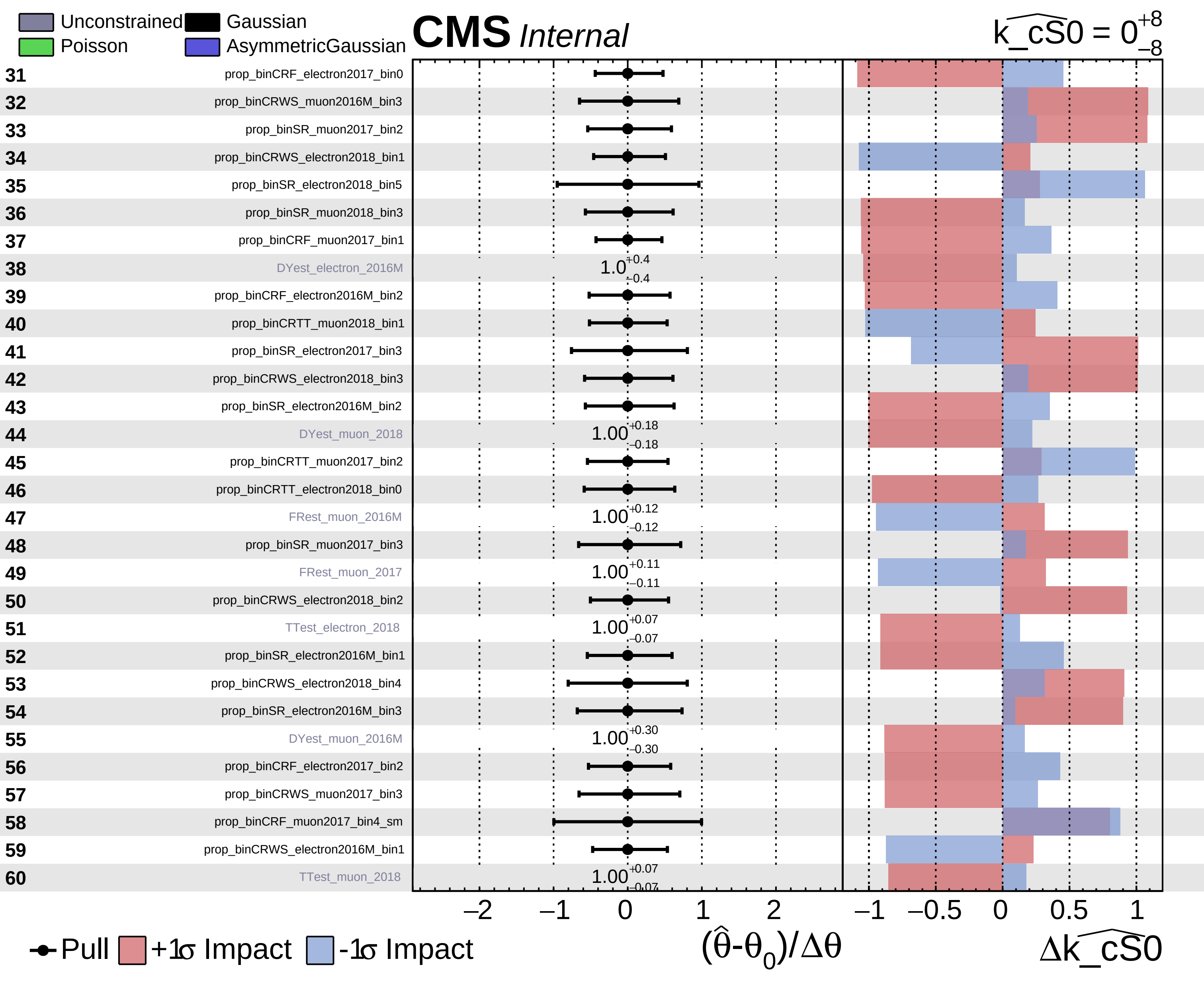
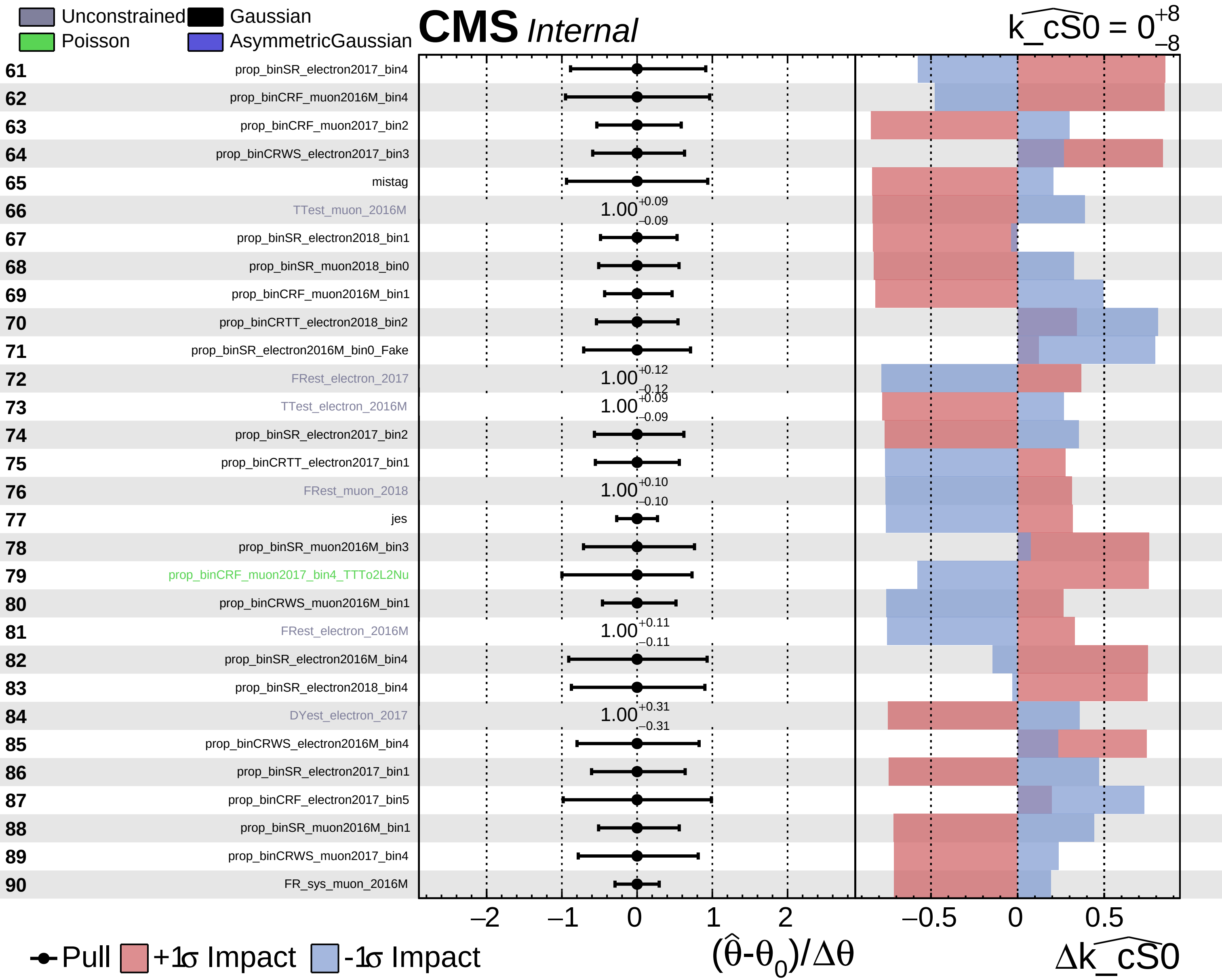


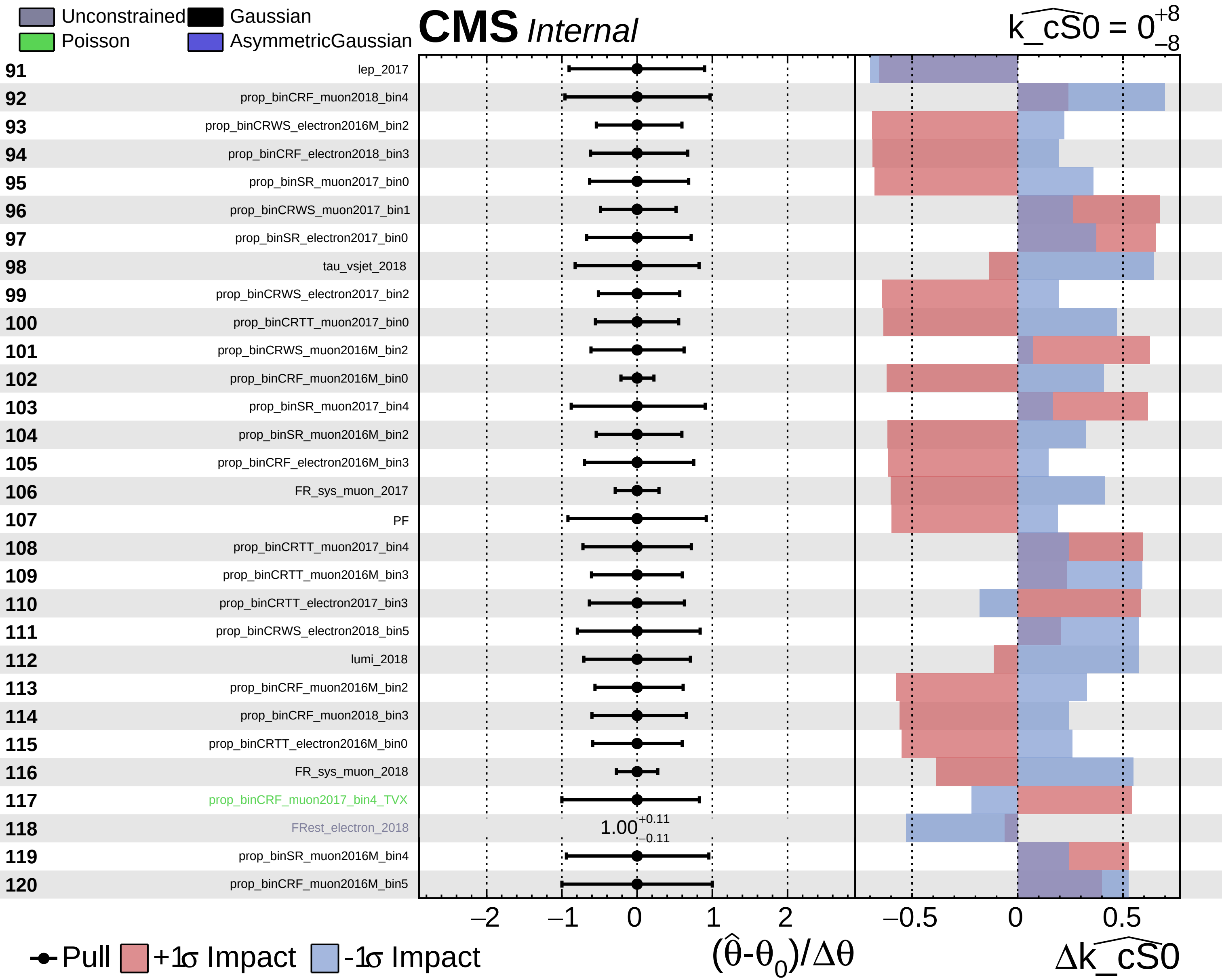
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = 0 +8 −8
1 prop_binSR_muon2018_bin5
2 prop_binSR_muon2017_bin5
3 prop_binSR_electron2016M_bin5
4 prop_binSR_electron2017_bin5
5 prop_binCRTT_muon2018_bin0
6 prop_binCRF_muon2018_bin0
7 prop_binCRF_electron2018_bin0
8 prop_binCRF_electron2018_bin1
9 pdf_total
10 prop_binCRF_electron2018_bin2
11 prop_binSR_muon2018_bin1
12 prop_binSR_electron2018_bin2
13 DYest_muon_2017 1.00 +0.24 −0.24
14 prop_binSR_muon2016M_bin5
15 prop_binCRF_muon2018_bin2
16 prop_binSR_muon2018_bin2
17 TTest_muon_2017 1.00 +0.07 −0.07
18 prop_binCRTT_electron2017_bin0
19 TTest_electron_2017 1.00 +0.07 −0.07
20 prop_binSR_electron2018_bin3
21 DYest_electron_2018 1.0 +0.4 −0.4
22 prop_binCRF_electron2016M_bin0
23 prop_binCRF_muon2017_bin0
24 prop_binCRTT_muon2016M_bin0
25 prop_binCRF_electron2016M_bin1
26 prop_binCRTT_muon2018_bin2
27 prop_binCRTT_electron2017_bin2
28 prop_binSR_muon2017_bin1
29 prop_binCRF_electron2017_bin1
30 prop_binCRF_muon2018_bin1
−2 −1 0 1 2
−2 0 2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS0



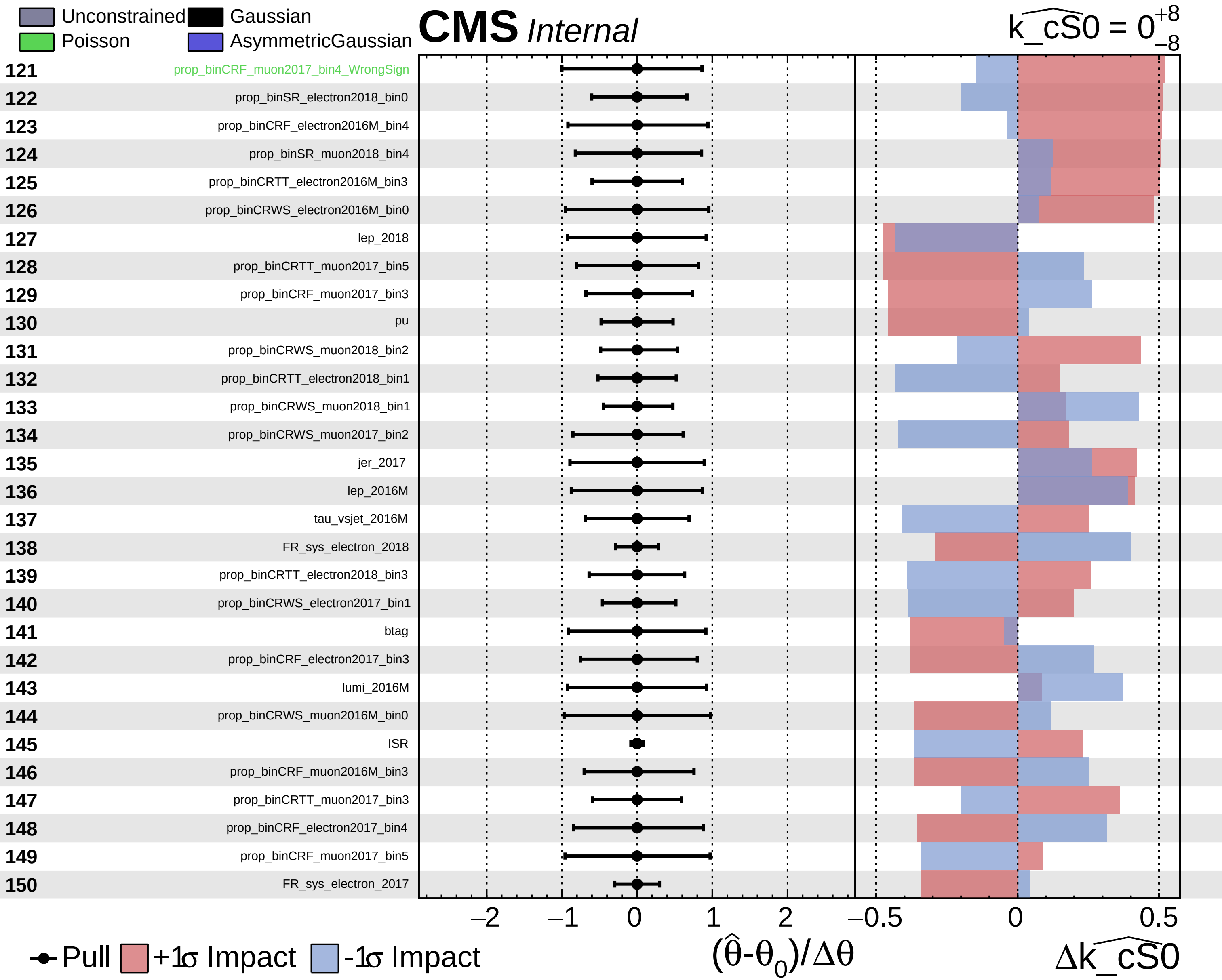
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS0} = 0^{+8}_{-8}$
31 prop_binCRF_electron2017_bin0
32 prop_binCRWS_muon2016M_bin3
33 prop_binSR_muon2017_bin2
34 prop_binCRWS_electron2018_bin1
35 prop_binSR_electron2018_bin5
36 prop_binSR_muon2018_bin3
37 prop_binCRF_muon2017_bin1
38 DYest_electron_2016M $1.0^{+0.4}_{-0.4}$
39 prop_binCRF_electron2016M_bin2
40 prop_binCRTT_muon2018_bin1
41 prop_binSR_electron2017_bin3
42 prop_binCRWS_electron2018_bin3
43 prop_binSR_electron2016M_bin2
44 DYest_muon_2018 $1.00^{+0.18}_{-0.18}$
45 prop_binCRTT_muon2017_bin2
46 prop_binCRTT_electron2018_bin0
47 FRest_muon_2016M $1.00^{+0.12}_{-0.12}$
48 prop_binSR_muon2017_bin3
49 FRest_muon_2017 $1.00^{+0.11}_{-0.11}$
50 prop_binCRWS_electron2018_bin2
51 TTest_electron_2018 $1.00^{+0.07}_{-0.07}$
52 prop_binSR_electron2016M_bin1
53 prop_binCRWS_electron2018_bin4
54 prop_binSR_electron2016M_bin3
55 DYest_muon_2016M $1.00^{+0.30}_{-0.30}$
56 prop_binCRF_electron2017_bin2
57 prop_binCRWS_muon2017_bin3
58 prop_binCRF_muon2017_bin4_sm
59 prop_binCRWS_electron2016M_bin1
60 TTest_muon_2018 $1.00^{+0.07}_{-0.07}$
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS0}$



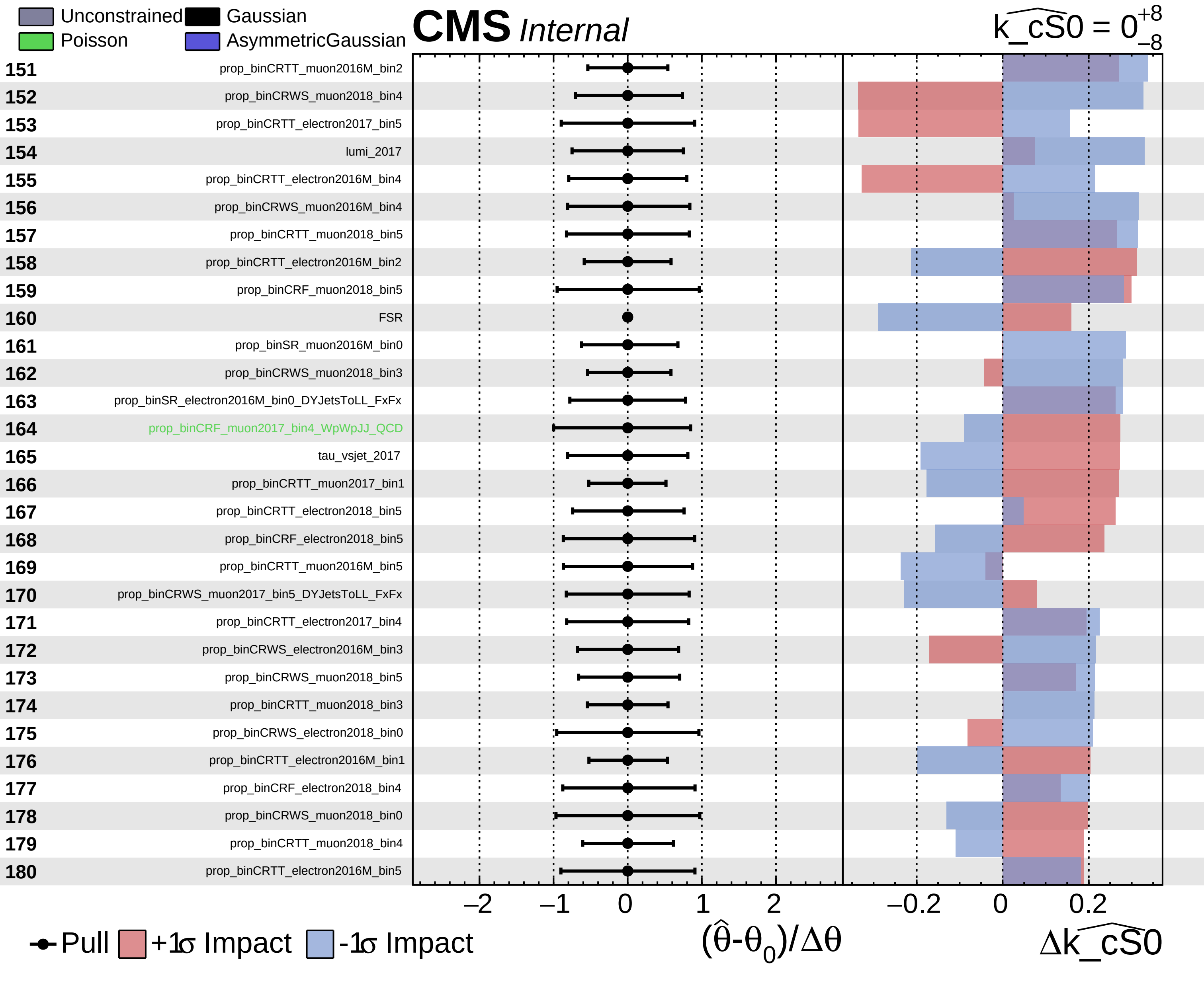
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = 0+8−8
61 prop_binSR_electron2017_bin4
62 prop_binCRF_muon2016M_bin4
63 prop_binCRF_muon2017_bin2
64 prop_binCRWS_electron2017_bin3
65 mistag
66 TTest_muon_2016M 1.00+0.09−0.09
67 prop_binSR_electron2018_bin1
68 prop_binSR_muon2018_bin0
69 prop_binCRF_muon2016M_bin1
70 prop_binCRTT_electron2018_bin2
71 prop_binSR_electron2016M_bin0_Fake
72 FRest_electron_2017 1.00+0.12−0.12
73 TTest_electron_2016M 1.00+0.09−0.09
74 prop_binSR_electron2017_bin2
75 prop_binCRTT_electron2017_bin1
76 FRest_muon_2018 1.00+0.10−0.10
77 jes
78 prop_binSR_muon2016M_bin3
79 prop_binCRF_muon2017_bin4_TTTo2L2Nu
80 prop_binCRWS_muon2016M_bin1
81 FRest_electron_2016M 1.00+0.11−0.11
82 prop_binSR_electron2016M_bin4
83 prop_binSR_electron2018_bin4
84 DYest_electron_2017 1.00+0.31−0.31
85 prop_binCRWS_electron2016M_bin4
86 prop_binSR_electron2017_bin1
87 prop_binCRF_electron2017_bin5
88 prop_binSR_muon2016M_bin1
89 prop_binCRWS_muon2017_bin4
90 FR_sys_muon_2016M
−2 −1 0 1 2
−0.5 0 0.5
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cS0



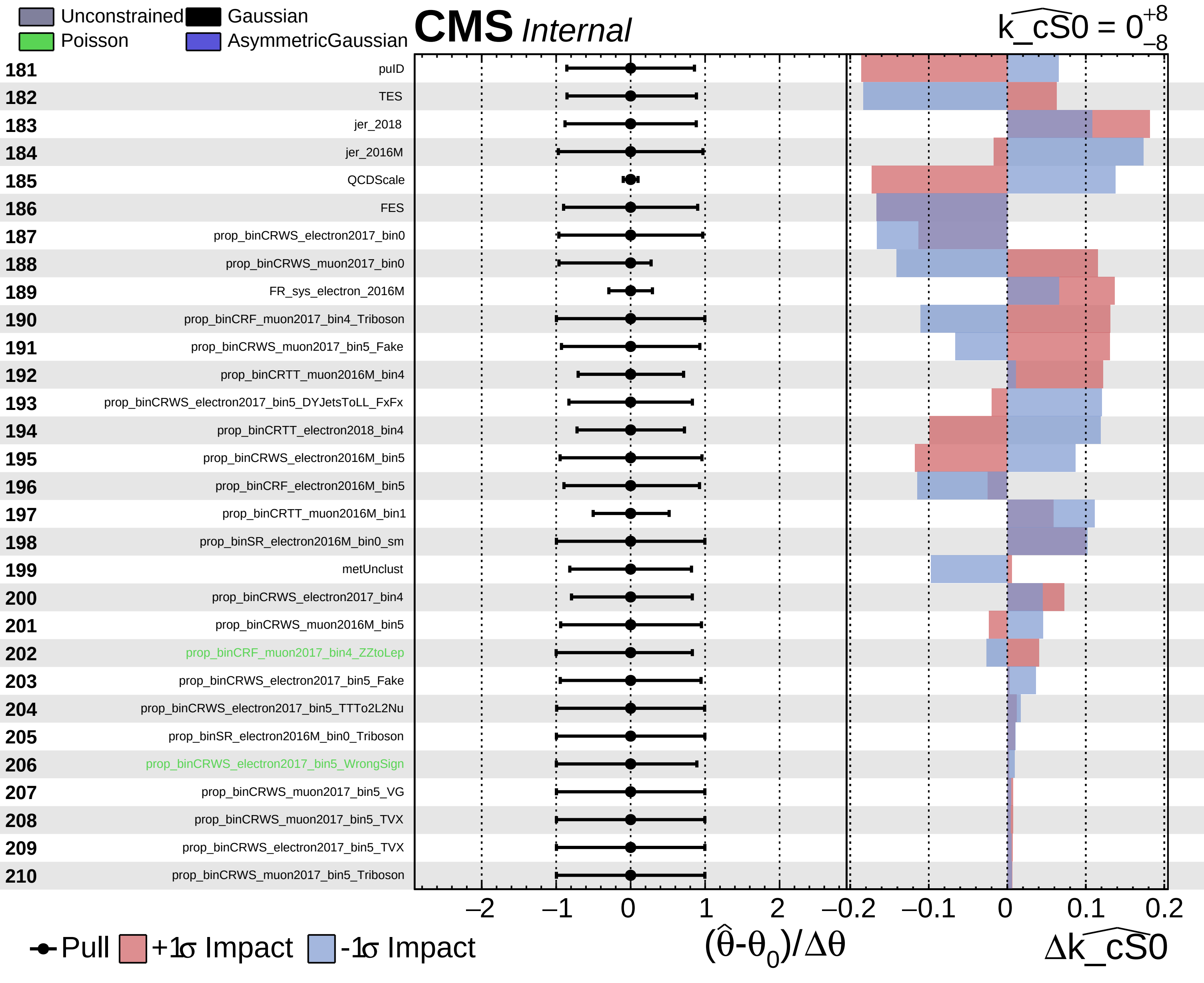
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = 0+8−8
91 lep_2017
92 prop_binCRF_muon2018_bin4
93 prop_binCRWS_electron2016M_bin2
94 prop_binCRF_electron2018_bin3
95 prop_binSR_muon2017_bin0
96 prop_binCRWS_muon2017_bin1
97 prop_binSR_electron2017_bin0
98 tau_vsjet_2018
99 prop_binCRWS_electron2017_bin2
100 prop_binCRTT_muon2017_bin0
101 prop_binCRWS_muon2016M_bin2
102 prop_binCRF_muon2016M_bin0
103 prop_binSR_muon2017_bin4
104 prop_binSR_muon2016M_bin2
105 prop_binCRF_electron2016M_bin3
106 FR_sys_muon_2017
107 PF
108 prop_binCRTT_muon2017_bin4
109 prop_binCRTT_muon2016M_bin3
110 prop_binCRTT_electron2017_bin3
111 prop_binCRWS_electron2018_bin5
112 lumi_2018
113 prop_binCRF_muon2016M_bin2
114 prop_binCRF_muon2018_bin3
115 prop_binCRTT_electron2016M_bin0
116 FR_sys_muon_2018
117 prop_binCRF_muon2017_bin4_TVX
118 FRest_electron_2018
1.00+0.11−0.11
119 prop_binSR_muon2016M_bin4
120 prop_binCRF_muon2016M_bin5
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS0

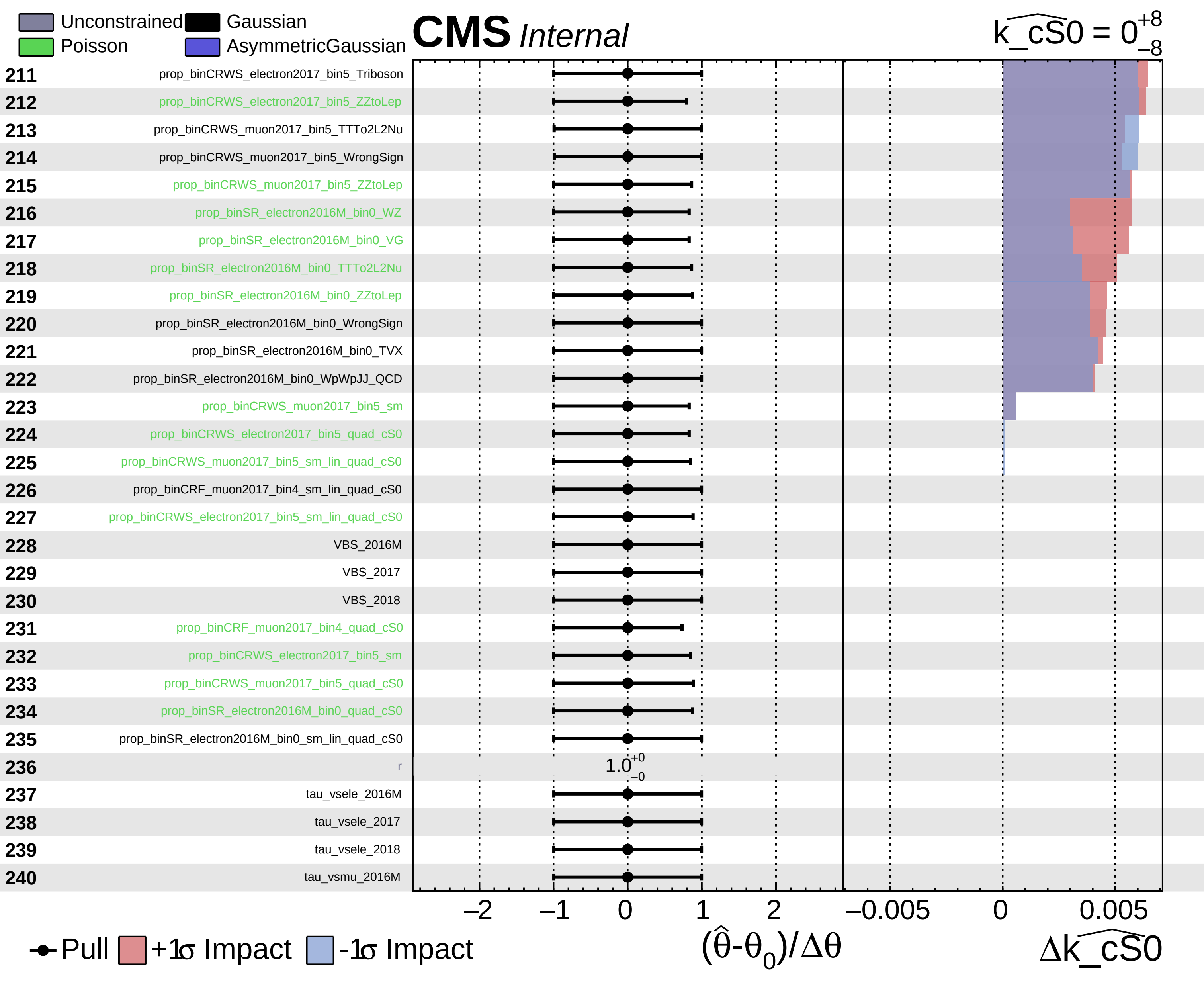


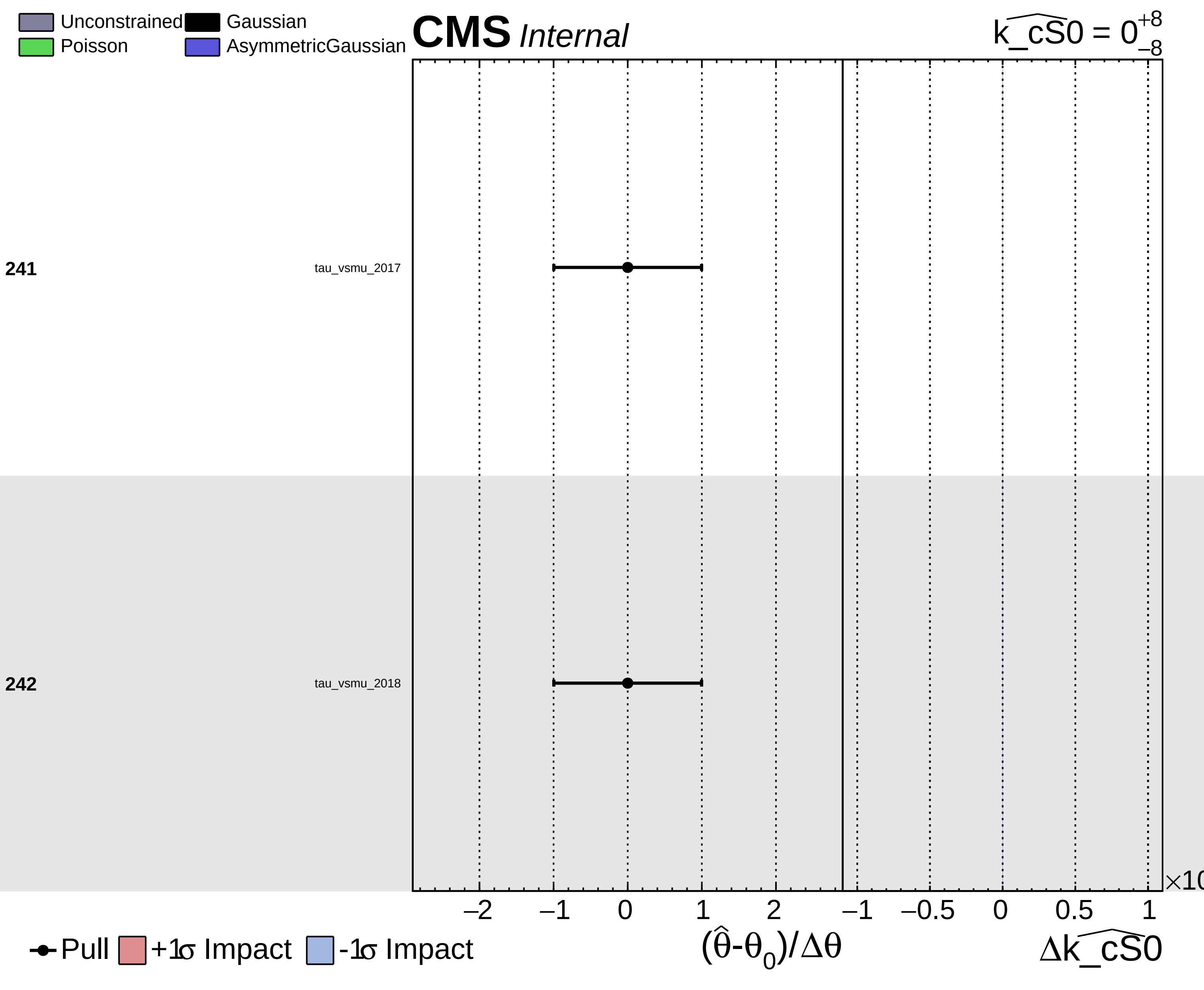
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = 0^{+8}_{−8}
121 prop_binCRF_muon2017_bin4_WrongSign
122 prop_binSR_electron2018_bin0
123 prop_binCRF_electron2016M_bin4
124 prop_binSR_muon2018_bin4
125 prop_binCRTT_electron2016M_bin3
126 prop_binCRWS_electron2016M_bin0
127 lep_2018
128 prop_binCRTT_muon2017_bin5
129 prop_binCRF_muon2017_bin3
130 pu
131 prop_binCRWS_muon2018_bin2
132 prop_binCRTT_electron2018_bin1
133 prop_binCRWS_muon2018_bin1
134 prop_binCRWS_muon2017_bin2
135 jer_2017
136 lep_2016M
137 tau_vsjet_2016M
138 FR_sys_electron_2018
139 prop_binCRTT_electron2018_bin3
140 prop_binCRWS_electron2017_bin1
141 btag
142 prop_binCRF_electron2017_bin3
143 lumi_2016M
144 prop_binCRWS_muon2016M_bin0
145 ISR
146 prop_binCRF_muon2016M_bin3
147 prop_binCRTT_muon2017_bin3
148 prop_binCRF_electron2017_bin4
149 prop_binCRF_muon2017_bin5
150 FR_sys_electron_2017
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS0} = 0^{+8}_{-8}$
151 prop_binCRTT_muon2016M_bin2
152 prop_binCRWS_muon2018_bin4
153 prop_binCRTT_electron2017_bin5
154 lumi_2017
155 prop_binCRTT_electron2016M_bin4
156 prop_binCRWS_muon2016M_bin4
157 prop_binCRTT_muon2018_bin5
158 prop_binCRTT_electron2016M_bin2
159 prop_binCRF_muon2018_bin5
160 FSR
161 prop_binSR_muon2016M_bin0
162 prop_binCRWS_muon2018_bin3
163 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
164 prop_binCRF_muon2017_bin4_WpWpJJ_QCD
165 tau_vsjet_2017
166 prop_binCRTT_muon2017_bin1
167 prop_binCRTT_electron2018_bin5
168 prop_binCRF_electron2018_bin5
169 prop_binCRTT_muon2016M_bin5
170 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
171 prop_binCRTT_electron2017_bin4
172 prop_binCRWS_electron2016M_bin3
173 prop_binCRWS_muon2018_bin5
174 prop_binCRTT_muon2018_bin3
175 prop_binCRWS_electron2018_bin0
176 prop_binCRTT_electron2016M_bin1
177 prop_binCRF_electron2018_bin4
178 prop_binCRWS_muon2018_bin0
179 prop_binCRTT_muon2018_bin4
180 prop_binCRTT_electron2016M_bin5
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS0}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = 0 +8 −8
181 puID
182 TES
183 jer_2018
184 jer_2016M
185 QCDScale
186 FES
187 prop_binCRWS_electron2017_bin0
188 prop_binCRWS_muon2017_bin0
189 FR_sys_electron_2016M
190 prop_binCRF_muon2017_bin4_Triboson
191 prop_binCRWS_muon2017_bin5_Fake
192 prop_binCRTT_muon2016M_bin4
193 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
194 prop_binCRTT_electron2018_bin4
195 prop_binCRWS_electron2016M_bin5
196 prop_binCRF_electron2016M_bin5
197 prop_binCRTT_muon2016M_bin1
198 prop_binSR_electron2016M_bin0_sm
199 metUnclust
200 prop_binCRWS_electron2017_bin4
201 prop_binCRWS_muon2016M_bin5
202 prop_binCRF_muon2017_bin4_ZZtoLep
203 prop_binCRWS_electron2017_bin5_Fake
204 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
205 prop_binSR_electron2016M_bin0_Triboson
206 prop_binCRWS_electron2017_bin5_WrongSign
207 prop_binCRWS_muon2017_bin5_VG
208 prop_binCRWS_muon2017_bin5_TVX
209 prop_binCRWS_electron2017_bin5_TVX
210 prop_binCRWS_muon2017_bin5_Triboson
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS0


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = 0+8−8
211 prop_binCRWS_electron2017_bin5_Triboson
212 prop_binCRWS_electron2017_bin5_ZZtoLep
213 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
214 prop_binCRWS_muon2017_bin5_WrongSign
215 prop_binCRWS_muon2017_bin5_ZZtoLep
216 prop_binSR_electron2016M_bin0_WZ
217 prop_binSR_electron2016M_bin0_VG
218 prop_binSR_electron2016M_bin0_TTTo2L2Nu
219 prop_binSR_electron2016M_bin0_ZZtoLep
220 prop_binSR_electron2016M_bin0_WrongSign
221 prop_binSR_electron2016M_bin0_TVX
222 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
223 prop_binCRWS_muon2017_bin5_sm
224 prop_binCRWS_electron2017_bin5_quad_cS0
225 prop_binCRWS_muon2017_bin5_sm_lin_quad_cS0
226 prop_binCRF_muon2017_bin4_sm_lin_quad_cS0
227 prop_binCRWS_electron2017_bin5_sm_lin_quad_cS0
228 VBS_2016M
229 VBS_2017
230 VBS_2018
231 prop_binCRF_muon2017_bin4_quad_cS0
232 prop_binCRWS_electron2017_bin5_sm
233 prop_binCRWS_muon2017_bin5_quad_cS0
234 prop_binSR_electron2016M_bin0_quad_cS0
235 prop_binSR_electron2016M_bin0_sm_lin_quad_cS0
236 r 1.0+0−0
237 tau_vsele_2016M
238 tau_vsele_2017
239 tau_vsele_2018
240 tau_vsmu_2016M
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS0


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS0 = 0 +8 −8
241
tau_vsmu_2017
242
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS0